

Shibo Jin

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EDUCATION

<i>University of Pennsylvania (UPenn), USA</i> <i>Doctor of Philosophy (PhD) – Computer & Information Science</i>	Sep. 2026 – May 2030 (expected)
<i>University of Pennsylvania (UPenn), USA</i> <i>Master of Science in Engineering – Computer & Information Science</i>	Sep. 2024 – May 2026
<i>University of British Columbia (UBC), Canada</i> <i>Bachelor of Applied Science - Computer Engineering</i>	Sept. 2020 – May 2024

RESEARCH INTERESTS

My research focuses on LLM agents and multi-agent systems, particularly how agents create, solve, and learn from challenging tasks. I am interested in self-play, self-evolving agents, agent-generated problems and interactive environments, and evaluation methods that measure both task-solving and task-design capabilities. I also study how interactions among agents can produce useful training curricula and drive the co-evolution of reasoning and decision-making abilities.

INTERNSHIP EXPERIENCE

<i>AI Data Platform Intern, ByteDance, Beijing</i>	Jun. /2025 – Jun. /2025
- Developed a data scoring algorithm to evaluate the training effectiveness of individual samples, enabling targeted data selection for efficient large language model training. - Implemented a lightweight scoring model and integrated it into the data pipeline to prioritize high-value samples, improving model performance with reduced data selection speed and training data size. - Conducted comparative benchmark experiments to evaluate the effectiveness of the data scoring algorithm in improving model performance and verified the model's generalization capability across varying data scales.	
<i>Fabric Shell (C+AI) SWE Intern, Microsoft, Beijing</i>	May. /2024 – Aug. /2024
- Integrated Copilot into the Task Flow workspace platform, enhancing its functionality and user experience. - Developed and implemented prompt tuning, embedded retrieval, and various task flow related skills to improve Copilot's adaptability and effectiveness within the Task Flow platform. - Designed and deployed extensive tests, making necessary adjustments to ensure Copilot's robust and reliable performance when addressing user queries.	
<i>AI Technology Assistant, Lenovo, Beijing</i>	Jul. /2023 – Aug. /2023
- Collaborated in creating a fund management platform for limited partners and corporation's information disclosure. - Applied AI analytics to forecast client funds and improve data flow and user experience. - Gained a comprehensive understanding of the project lifecycle, emphasizing AI-based strategies and implementations.	

PUBLICATIONS

Selected Papers (* indicates equal contribution)

- (Under review) **Shibo Jin***, Zhiqiu Xu*, Yiling Zou, Shangyu Gong, Mayur Naik. "Environment Duels: A Microcosm of LLM Co-Evolution."
- (COLM 2026) Zhiqiu Xu*, **Shibo Jin***, Shreya Arya, Mayur Naik. "MathDuels: A Self-Play Benchmark That Grows." In COLM 2026. [pdf]
- (CVPR 2026) Zhiwei Zheng, **Shibo Jin**, Lingjie Liu, Mingmin Zhao. "Next-Scale Autoregressive Models for Text-to-Motion Generation." In CVPR 2026. [pdf]

RESEARCH EXPERIENCE

Environment Duels

- Co-developed an agentic benchmark in which LLMs author interactive environments with privileged hints and solve one another's tasks.
- Evaluated solver success, environment difficulty, and hint lift across nine frontier models.
- Studied how these measures relate to reinforcement-learning gains.

MathDuels

- Designed and implemented a multi-LLM self-play protocol for adversarial math problem generation and solving.
- Built automated answer extraction and symbolic & numeric verification; evaluated author and solver abilities across 19 frontier models.